

SIMATS ENGINEERING

Department of ComputerScience and Engineering

ASSESSMENT TOOL2 - CONCEPT MAPPING

Course Code:	CSA1205	Course Title:	Computer Architecture for Multicore Systems
CO Assessed:	CO1	Assessment:	ASSESSMENT TOOL 2- CONCEPT MAPPING
Weightage:	35%	Total Marks:	25
CO1 Statement:	Analyze the functional units of a computer system including CPU, memory, bus structures, and I/O components by examining instruction execution cycles, addressing modes, and performance metrics such as CPI, clock cycles, and benchmarking (BL4)		
Student Name:		Reg.No.:	
Department:		Date:	

Instruction to Students

1. Answer two questions as per the Instruction.
2. Each question carries 12.5 mark.
3. Only the appropriate Tools can be used
4. Time allotted for the exam is 60 min

ANNEXURE A

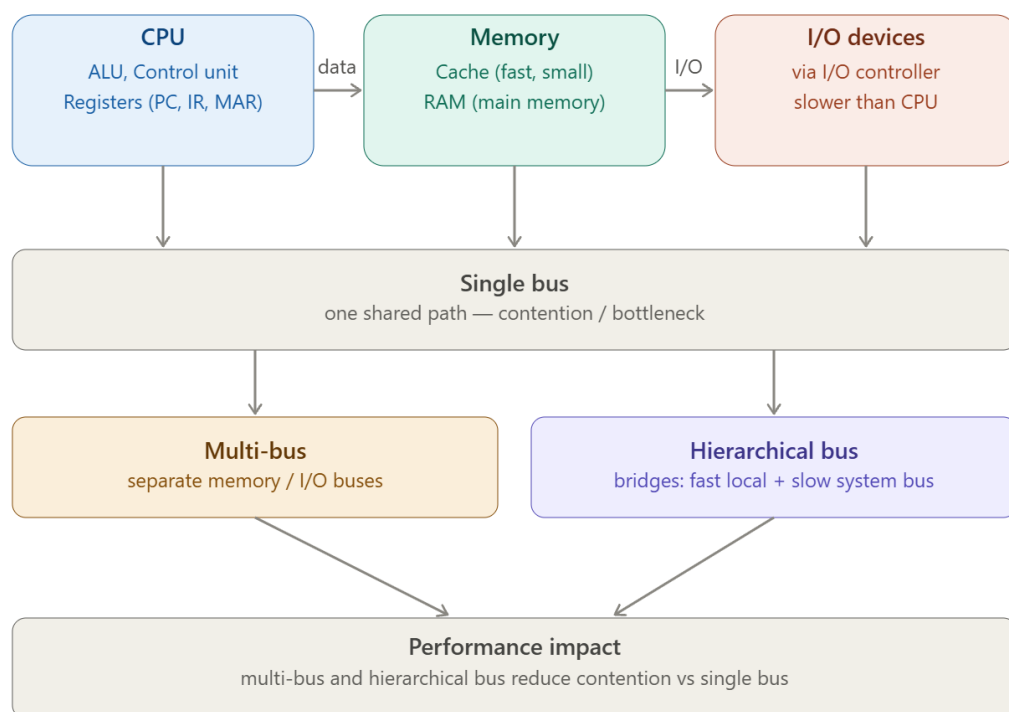
1. **Construct a detailed concept map showing the relationship between CPU (ALU, Control Unit, Registers), memory (cache, RAM), and I/O devices. Extend the map to include single-bus, multi-bus, and hierarchical bus structures, and clearly illustrate how data and control signals flow among these components during execution. Indicate possible points of delay or bottlenecks and how different bus structures help in improving performance. Concept Identification (25%) — make sure every one of these appears as a labeled node:**

Answer:

- CPU: ALU, Control Unit, Registers (PC, IR, MAR, MBR, GPRs)
- Memory: Cache (L1/L2), RAM
- I/O devices and I/O controllers
- Bus types: Single-bus, Multi-bus, Hierarchical bus
- Bus lines: Data bus, Address bus, Control bus
- Performance concepts: bus contention/bottleneck, DMA

CPU (ALU + Control Unit + Registers) ↔ Memory: address/data exchanged over address bus and data bus; timing regulated by control bus signals from the Control Unit.

- Cache sits between CPU and RAM — a cache hit avoids a slow RAM access; a cache miss forces a RAM fetch, adding delay.
- I/O devices connect through an I/O controller onto the bus; since I/O is slower than the CPU, this is a common source of wait states unless DMA is used to bypass CPU involvement.
- Single-bus structure: one shared path for CPU, memory, and I/O → simple but creates contention when multiple components need the bus at once (bottleneck).
- Multi-bus structure: separate buses (e.g., memory bus, I/O bus) → reduces contention, allows more parallel transfers.
- Hierarchical bus: bridges connect a fast local bus (CPU–cache) to slower system/expansion buses → isolates slow devices from fast ones, best scalability and performance.



2. **Develop a comprehensive concept map linking the instruction execution cycle (fetch, decode, execute) with CPU performance parameters such as CPI, clock cycles, and execution time. Show how each stage contributes to overall execution time, and include connections to factors like memory access, instruction complexity, and pipeline stalls. Also, represent how benchmarking is used to evaluate performance.**

Answer:

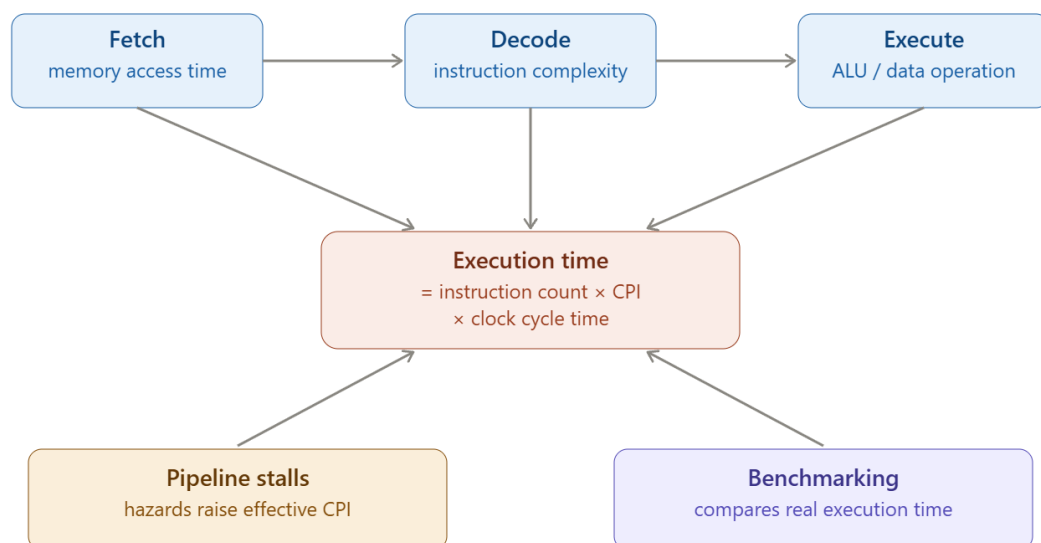
Concept Identification Cycle stages: Fetch, Decode, Execute

- Metrics: CPI (cycles per instruction), Clock cycle time, Instruction count
- Formula: Execution time = Instruction count $\times$ CPI $\times$ Clock cycle time
- Influencing factors: memory access time, instruction complexity, pipeline stalls/hazards
- Evaluation: Benchmarking

Linkages

Fetch retrieves the instruction from memory — its speed depends directly on memory access time (cache hit vs. RAM miss).

- Decode interprets the opcode/operands — more complex instructions or addressing modes take more decode cycles.
- Execute performs the actual ALU/data operation, consuming clock cycles that feed into CPI.
- CPI, instruction count, and clock cycle time combine into the execution-time formula — this is the node that ties the whole map together.
- Pipeline stalls (structural, data, control hazards) increase effective CPI, directly worsening execution time.
- Benchmarking measures real execution time across these factors to compare CPUs objectively rather than relying on clock speed alone.



Code of Conduct:

I G.Saranraj(192511182) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

MARKS ALLOCATION

Criteria	Concept Identification (25%)	Linkages (25%)	Organization (20%)	Integration of Literature (20%)	Clarity (10%)
Q1					
Q2					

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand